

Quarterly Mechanical Inspection Report INSP-2024-0315

P-101 — Boiler Feed Water Pump P-101 | Inspected 2024-03-15 by Priya Sharma

Report number	INSP-2024-0315	Inspection date	2024-03-15
Equipment	P-101 (Sulzer CP 150-400)	Criticality	Class A
Inspector	Priya Sharma, Reliability Engineer	Witness	Rajesh Kumar
Running hours	63,480 h since 2016 commissioning	Next inspection due	2024-06-14

1. Scope and method

Quarterly condition assessment of the non-drive-end and drive-end bearings, lubrication system, mechanical seal and coupling alignment, carried out with the machine running at steady duty and again during the 90-minute shutdown window granted by operations. Wear assessment by outer-race shell-thickness comparison against the OEM Appendix D reference set; vibration by portable collector with 1x/2x spectral capture; oil condition by laboratory analysis of a sample drawn from the bearing return line.

2. Findings

- NDE bearing wear measured at 71% of the OEM replacement limit by shell-thickness comparison.
- Overall vibration 5.8 mm/s RMS at the NDE bearing housing, against a 7.1 mm/s alarm limit.
- NDE bearing housing temperature 74 degC at steady 148 m3/h duty.
- Lube oil header pressure 1.9 bar, within limits but 0.2 bar below the commissioning value.
- Oil sample: ISO 4406 code 18/16/13, iron content 22 ppm, up from 9 ppm in September 2023.
- Wear pattern similar to 2022 failure: identical scoring on the NDE outer race at the load zone, and the same 1x/2x running-speed spectral signature that preceded the August 2022 seizure.

3. Measurements against limits

Parameter	This inspection	Previous (2023-09-19)	OEM limit	Assessment
NDE bearing wear (%)	71	52	85	Elevated — trend accelerating
Overall vibration (mm/s RMS)	5.8	4.4	7.1	Within alarm, rising
NDE housing temperature (degC)	74	68	95	Within limit
Lube oil header pressure (bar)	1.9	2.0	1.4 min	Within limit, drifting down
Oil iron content (ppm)	22	9	20 alarm	Above alarm

4. Wear pattern assessment

Wear pattern similar to 2022 failure: identical scoring on the NDE outer race at the load zone, and the same 1x/2x running-speed spectral signature that preceded the August 2022 seizure.

The scoring on the outer race load zone and the 1x/2x spectral signature are the same pattern recorded before the August 2022 seizure of this machine. On the OEM's own criteria (Section 7.4) a wear pattern matching a previous seizure signature is to be treated as pre-replacement irrespective of the measured percentage.

5. Bearing wear trend since the 2022 replacement

Date	Wear (%)	Percent of OEM limit	Source
2022-03-10	62	73%	Maintenance_Log
2022-06-15	84	99%	Maintenance_Log
2022-08-26	0	0%	Maintenance_Log
2023-03-14	34	40%	Maintenance_Log
2023-09-19	52	61%	Maintenance_Log
2024-03-15	71	84%	Inspection_2024_0315

Wear has advanced roughly 7 percentage points per quarter since the bearing set was renewed on 2022-08-26. On that gradient the 85% replacement limit is reached in September 2024.

6. Conclusion and recommendation

- The machine is fit for continued service to the next quarterly inspection, conditional on the recommendations below being actioned.
- Plan the bearing replacement now rather than on reaching the limit: the spare set has a six-week lead time and the last replacement took a 240-minute outage window.
- Restore quarterly lube oil analysis and investigate the iron content rise from 9 ppm to 22 ppm, which on this machine previously indicated auxiliary lube pump wear.
- Plot bearing wear against the OEM limit on every subsequent report — corrective action 5 of RCA-2022-0825 requires it.

Distribution: D. Krishnan (Maintenance Manager), Rajesh Kumar (Senior Technician), S. Ramaswamy (Operations Foreman). Filed against P-101 in the asset register.